

NAVANEETA PADMAKUMAR

Richardson, Texas

+1 945-338-9465 | nxp230016@utdallas.edu | [linkedin.com/in/navaneetapk](https://www.linkedin.com/in/navaneetapk) | github.com/nxp0999

Education

The University of Texas at Dallas

Expected May 2026

Master of Science in Computer Science - Data Science

Richardson, TX

- CGPA: 3.6 / 4.0
- Relevant coursework: Machine Learning, Statistics for Data Science, Database Design, Artificial Intelligence

Government Engineering College, Trivandrum

July 2021

Bachelor of Technology (Honors) in Electronics and Communication

Trivandrum, India

- GPA: 8.87 / 10 — Chairperson – IEEE WiE, Under Secretary General – MUN

Professional Certifications

Databricks Certified Data Engineer Professional – January 2025 (Valid through January 2027)

PG Advanced Program in Data Science & Machine Learning – IIT Madras – November 2024

Technical Skills

Languages: Python, R, SQL, Java, C, Node.js

Data Science & ML: scikit-learn, PyTorch, scikit-learn, TensorFlow, NumPy, Pandas, SciPy, Feature Engineering

Statistical Analysis: Regression, Classification, Clustering, A/B Testing, Time Series, Hypothesis Testing

Visualization & Tools: Matplotlib, Seaborn, Git, R Studio, HuggingFace

Data Engineering: MySQL, MongoDB, Docker, Kubernetes, Shell Scripting,

Technologies: Linux, MATLAB, VS Code, Databricks, Apache Spark

Data Science Project Experience

GlobalRides – Predictive Analytics & Database System | SQL, Python, MySQL

January 2025 – May 2025

- Built predictive models using historical order data to forecast demand patterns across 12 restaurants and 11 ride routes, improving resource allocation efficiency by 23% through time-series analysis and regression modeling
- Analyzed 260+ orders to identify customer segmentation patterns and optimize pricing strategy, discovering that power users (15+ orders) generated \$1,694 revenue with 4.2/5.0 satisfaction through statistical profiling
- Designed relational database with 26 normalized tables and indexed queries, processing complex joins for real-time analytics on concurrent operations; achieved 95%+ order fulfillment within SLA windows

CSMC Mentoring Platform – Queuing Analysis & Optimization | C, Statistical Modeling

September 2024

- Applied queuing theory and statistical analysis to optimize tutor-student matching algorithm, reducing maximum wait time variance to < 800 microseconds through probability distribution modeling and performance metrics tracking
- Analyzed system performance under 100+ concurrent users using Monte Carlo simulation, achieving 200-microsecond average response times through thread-safe queue optimization with O(1) complexity
- Designed A/B testing framework to evaluate scheduling algorithms, measuring fairness metrics and throughput improvements; documented statistical significance using hypothesis testing

Satellite Cluster Optimization via ML | Python, MATLAB, NumPy, Optimization

August 2020 – June 2021

- Improved satellite formation efficiency by 50% through machine learning-based trajectory optimization using Clohessy-Wiltshire equations; reduced fuel consumption via gradient descent and parametric analysis of orbital dynamics
- Developed automated numerical methods and visualization tools in Python/MATLAB for rapid hyperparameter tuning and control optimization; created interactive dashboards for algorithm performance comparison and sensitivity analysis
- Validated predictive models against simulation data achieving 92% accuracy in trajectory forecasting, enabling cost-effective mission planning in resource-constrained space environments

Professional Experience

Datanimbus Technologies

July 2021 – January 2024

Platform and DevOps Engineer | Python, Docker, Kubernetes, MongoDB, APIs

Bangalore, India

- Automated CI/CD analytics pipeline using Python and shell scripting to track deployment metrics, test coverage, and build performance; reduced deployment time by 80% (hours to minutes) through data-driven optimization
- Analyzed system performance data across 20+ deployments to maintain 100% uptime, identifying bottlenecks and failure patterns through log analysis and anomaly detection; mentored by CTO on data-driven decision making
- Conducted exploratory data analysis on 300+ UI/UX issues across 25 modules, using statistical methods to prioritize bug fixes by severity and impact; resolved 50%+ client-reported defects through root cause analysis
- Streamlined onboarding through data analysis of training metrics, reducing ramp-up time 50% (4 to 2 weeks); acquired 3 clients through data-driven product presentations showcasing performance insights

ThoughtLine Technologies Pvt. Ltd., Technopark

June 2018

Data Analytics Intern

Trivandrum, India

- Performed statistical analysis to identify high-margin product categories and seasonal demand trends, generating actionable insights for targeted promotional campaigns; built regression models to forecast inventory needs
- Designed and deployed interactive analytics dashboards using Google Sheets and Python visualization libraries, providing real-time KPIs on product performance, inventory turnover, and category profitability to stakeholders
- Implemented data validation rules and quality checks that eliminated pricing errors, reducing customer complaints by 40%; standardized product specifications through systematic data transformation